

MARVIS STEWART REGO

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PROFESSIONAL SUMMARY

ML Engineer with hands-on production experience in data preprocessing, model training, evaluation, deployment, and post-deployment monitoring on Azure Databricks. Completing an MSc in Data Science and Artificial Intelligence (GPA: 4.0) at Middlesex University Dubai. Proficient in Python, scikit-learn, TensorFlow, NLP, and end-to-end data pipelines. Experienced collaborating with cross-functional teams to translate business requirements into scalable AI solutions, with a strong interest in applied AI research and innovation.

TECHNICAL SKILLS

AI / ML: Supervised & Unsupervised Learning, Deep Learning, NLP, LLMs, Generative AI, RAG, Anomaly Detection, Feature Engineering

Languages & Libraries: Python (Pandas, NumPy, Scikit-learn, TensorFlow, PySpark), SQL, R

MLOps & Platforms: Azure Databricks, MLflow, Delta Lake, ML Observability, Automated Drift Detection, CI/CD, ETL Pipelines

Cloud & Tools: Azure, AWS (S3, CloudFront), GCP, Power BI, Git, GitHub

PROFESSIONAL EXPERIENCE

Global Hotel Alliance (GHA)

Dubai, UAE

Machine Learning Engineer

Jan 2026 - Present

- Built and deployed end-to-end supervised classification models for member activation, from data collection and feature engineering through pipeline automation and continuous post-deployment monitoring.
- Engineered an ML pipeline on Databricks using Databricks Feature Store with strict time-windowing to prevent data leakage; optimized hyperparameters reducing model runtime from 6 hours to 10 minutes without accuracy loss.
- Deployed batch inference scoring 2M+ members into 10 deciles with differentiated promotional offers, communicating results directly to marketing stakeholders to optimize campaign ROI.
- Engineered scalable PySpark ETL pipelines to ingest, clean, and transform multi-million row datasets across multiple sources, with data quality checks and validation at each stage.
- Designed a multi-table ML observability framework with automated drift detection, data quality checks, and performance alerting, reducing manual intervention by 30% across all production models.
- Developed an LLM-powered scoring system using NLP and text mining to extract labels from unstructured property descriptions, with robust JSON parsing and fallback mechanisms ensuring output quality.
- Developed a member activity segmentation pipeline aggregating behavioral data, applying rule-based segmentation across funnel positions, and logging inference metadata into Databricks catalog tables.
- Delivered behavioral clustering on 30M+ loyalty records enabling precision-targeted outreach; built an anomaly and fraud detection pipeline with automated alerting for real-time monitoring.

PERSONAL PROJECTS

Student Dropout Risk Prediction System

Python, Scikit-learn | 2025

- Trained Gradient Boosting and Logistic Regression models on unstructured student behavioral data, engineering attendance and behavioral ratio features to achieve 80% accuracy and project a 50% reduction in tuition loss.
- Designed the full ML workflow including data preprocessing, feature selection, model evaluation, and results visualization, producing actionable risk scores for early intervention.

DreamTrip - AI Travel Orchestration Platform

GPT-4o, RAG, AWS, React | 2026

- Built an LLM-powered travel platform using GPT-4o mini with real-time RAG via Tavily API, transforming natural language prompts into structured multi-day itineraries with geospatial visualization, deployed on AWS S3 and CloudFront.
- Implemented a quality control layer to validate structured outputs, detect incomplete model responses, and apply fallback logic, ensuring consistent itinerary generation across varied user inputs.

Career Recommendation System

RAG, Knowledge Graph, OpenAI Embeddings, Cohere Rerank, GPT-5.4 mini | 2026

- Built a RAG-based career guidance system that maps user profiles, skills, and degrees to role and Coursera course recommendations through a knowledge graph.
- Implemented clarification prompts, OpenAI embedding retrieval, and Cohere reranking to prioritize 5-6 relevant recommendations before GPT-5.4 mini responses.

EDUCATION

Middlesex University Dubai

Sept 2025 - June 2026

MSc in Data Science and Artificial Intelligence | GPA: 4.0

Coursework: Machine Learning, Generative AI, Data Engineering, Data Visualisation

Mangalore University

Sept 2022 - July 2025

Bachelor's in Computer Application (BCA)

CERTIFICATIONS

Databricks Academy Gen AI Certification (2026) | Introduction to Data Science - Cisco (Aug 2025)